Feasibility-Aware Quantum-Classical Benchmarking
of a HUBO Formulation for Occlusion-Aware UAV Trajectory Optimization

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SCOPE

We introduce a quantum-compatible HUBO benchmarkfor occlusion-aware UAV trajectory optimization and a **feasibility-aware methodology** for evaluating such models. Classical baselines are deliberate: current quantum hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment chart a path toward future D-Wave / QAOA hardware studies.

1. The Problem

UAV target tracking in cluttered space must **simultaneously**:

- avoid obstacles, and
- keep **line-of-sight** to a target (occlusion-free).

Masnavi *et al.* solved this as continuous multi-convex MPC. We recast the **same problem class** as a discrete higher-order binary problem for quantum solvers.

2. HUBO Formulation

Binary $x[t, v] = 1$ iff UAV at cell v, time t. Seven cost terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}$$

Occlusion (Bresenham line-of-sight):

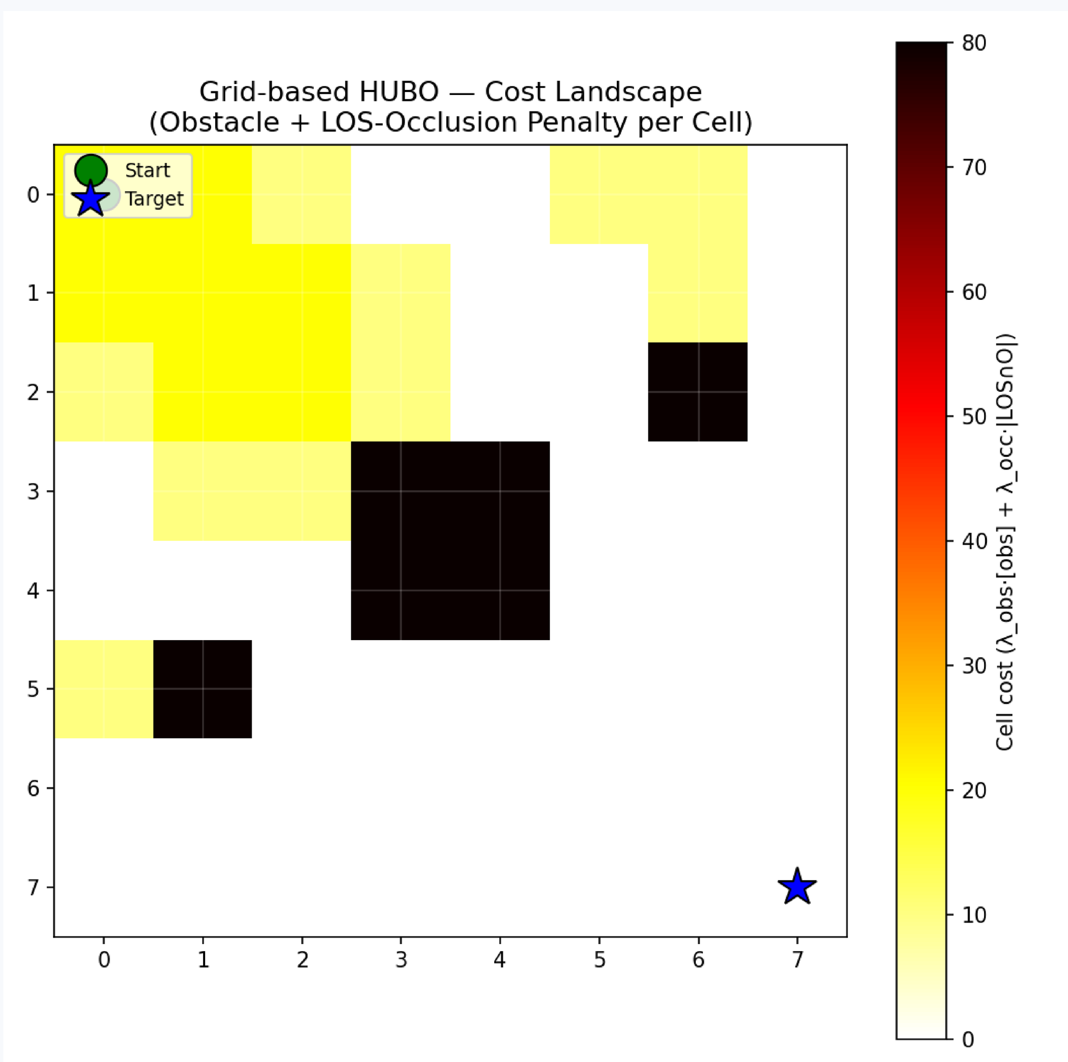
$$H_{\text{occ}} = \lambda_{\text{occ}} \Sigma_t \Sigma_u\, x[t, u]\, |\text{LOS}(u, \text{tgt}) \cap O|$$

Genuinely cubic proximity (lingering in a buffer cell):

$$H_{\text{prox}} = \lambda_{\text{prox}} \Sigma_{v_b} \Sigma_t\, x[t{-}1, v_b]x[t, v_b]x[t{+}1, v_b]$$

Why HUBO not QUBO? Quadratizing the cubic term needs ∼3,172 ancillas (∼30% more variables) with dense couplings that wreck sparsity.

Occlusion Cost Surface



Yellow = target occluded from that cell; dark = obstacle.

3. Feasibility-Aware Benchmarking

Raw HUBO energy **misleads** — annealers lower energy by *violating* soft constraints. Four safeguards:

(i) Energy decomposition
hard-penalty vs. soft-cost, kept separate

(ii) Independent constraint check
decode + verify, not energy thresholds

(iii) Matched sampling budgets
equal reads/runs across solvers

(iv) Penalty-weight sweep
exposes the constraint-exploitation regime

Two-Tier Design

State-vector QAOA scales as 2^n — impossible at n=960.

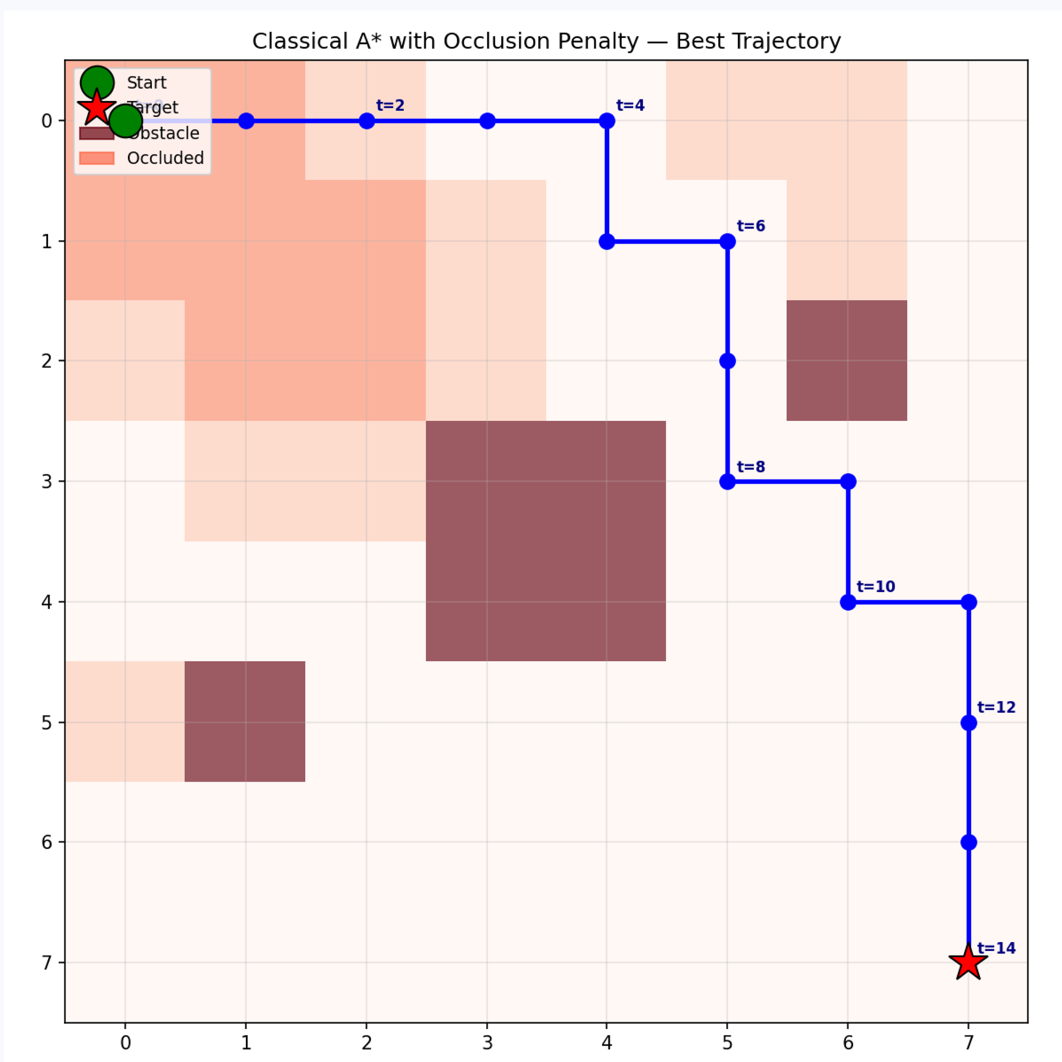
TIER 1	TIER 2
Full 8×8	Tiny 2×2
960 vars	8 vars
A* · SA · neal	+Qiskit QAOA

Classical A* Baseline

A* on the same grid with occlusion-weighted cell cost

$$1.0 + \lambda_{\text{occ}} |\text{LOS} \cap O| + \lambda_{\text{prox}} \mathbf{1}[v_b \in B]$$

grounds the HUBO against a conventional planner — answering whether the formulation captures **sensible planning** or is a benchmark artifact.



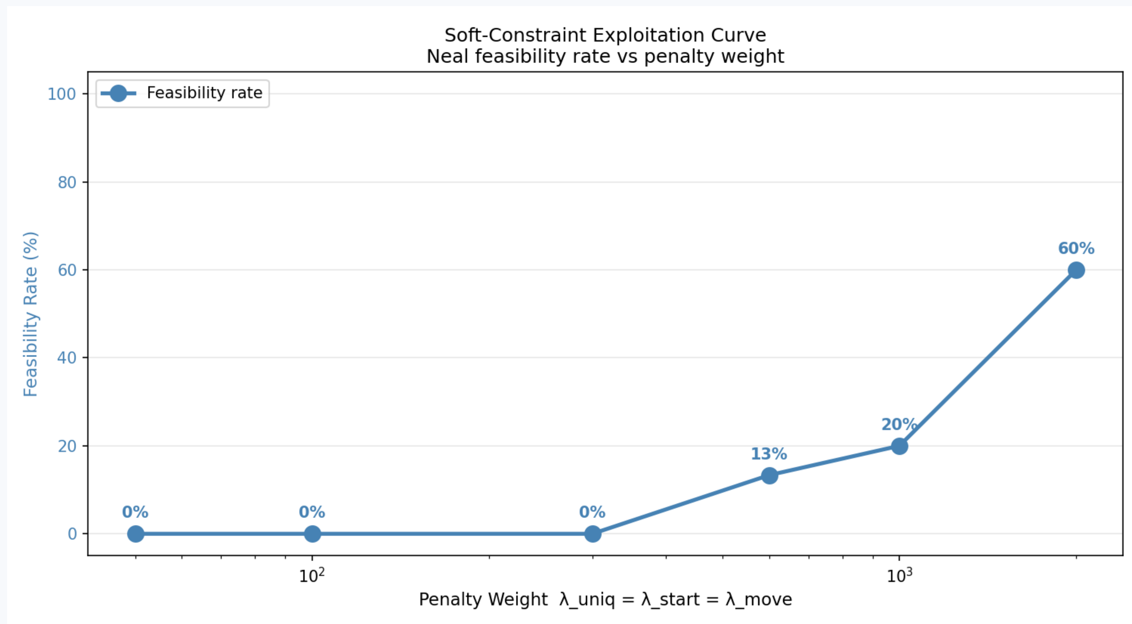
A* path: feasible, reaches target, 80% visibility.

4. Results: 8×8 Instance

Metric	A*	SA	neal
Soft cost	317	**303**	496
Feasibility	**100%**	100%	22%
Reaches target	**yes**	14%	8%
Visibility	**80%**	—	—
Runtime/sample	**10 ms**	47.8 s	0.26 s

A* and SA reach comparable quality; A* is ∼4,700× faster. neal is fast but only 22% feasible.

Headline: Constraint-Exploitation Curve



neal feasibility climbs **0%** → **60%** as λ: 50 → 2000

General-purpose annealers need penalty weights an **order of magnitude** above the objective scale to respect one-hot HUBO structure. A single λ would misreport feasibility.

Tier 2 & Takeaways

On 8 variables, **all four solvers** reach the same optimum ($E^* = -118.17$); QAOA is a correctness check (2,750× slower, no scaling).

The HUBO's value is not raw speed here, but compatibility with quantum solvers and extensibility to a **moving target**, where occlusion becomes intrinsically cubic — beyond A*'s shortest-path reach.

Future: moving target · real D-Wave hardware · vs. continuous MPC.

CONTACT + REFERENCES

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Selected references: [6] Masnavi et al., IEEE Access 2022; [7] Innan et al., QUAV, arXiv 2025; [8] Osaba et al., quantum drone routing, arXiv 2025; [9] Glover et al., 4OR 2019.